

Spectral Structure of the 30 Day SPX Implied Volatility Smile in Vendor Smoothed Surfaces

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<https://github.com/MichaelAafton/diffusion-modes-vol-surface>

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Abstract

I study the principal components of daily changes in the 30 day implied volatility smile of the S&P 500 index, using ten years of OptionMetrics volatility surfaces (2015 to 2024). A level factor carries 94.4% of the variance and a second factor 4.7%. The remaining modes fall off as a power law with exponent $p = 4.44$ (95% bootstrap interval 4.18 to 4.67). Simulated diffusion with a level factor, passed through the same panel construction, never exceeds $p = 2.63$, so a diffusing membrane observed directly cannot produce this tail. The surfaces are not observed directly, however. The vendor builds them with a Gaussian kernel, and when simulated diffusion is passed through that kernel its tail exponent falls between 4.2 and 4.5, inside the measured interval. The shape does not match: smoothed diffusion bends sharply downward on log scales while the measured spectrum stays close to a straight line. The second factor is partly tied to the seam between the separately smoothed call and put halves of the surface ($R^2 = 0.57$, interval 0.43 to 0.68), and several modes show negative autocorrelation over two to five days. With this instrument the question of which operator governs the smile cannot be settled. I describe the measurement that would settle it.

1 Introduction

The implied volatility smile of a liquid index changes every day, and the structure of those changes determines the risk of any portfolio of options. Principal component analysis of daily changes is the standard way to describe that structure. Cont and da Fonseca [1] found that a few factors, which can be read as level, slope and curvature, account for most of the daily variation of implied volatility surfaces.

A physical reading of the same decomposition treats the smile as an elastic membrane. Local shocks deform it, and an effective stiffness spreads and damps the deformation. Le Coz and Bouchaud [2] develop this kind of model for the forward rate curve. For the smile, the simplest version is a stochastic heat equation on a moneyness axis z ,

$$\partial_t p(z, t) = D \partial_z^2 p(z, t) + \xi(z, t), \quad (1)$$

where p is the deviation of implied volatility from its average shape and ξ is white noise. If this picture holds, the principal components of daily changes should look like the modes of Eq. (1) and their variances should fall off in a specific way with mode number.

This paper set out to test that prediction on SPX data. The test turned into a study of the instrument. The surfaces come from a vendor who smooths raw option prices with a kernel before publication, and that smoothing reshapes the spectrum enough to change the answer. The findings are:

1. The 30 day smile has a dominant level factor (94.4% of daily variance) and a second factor (4.7%), followed by a tail that falls as $k^{-4.44}$ over modes 2 to 7.
2. Diffusion observed without smoothing is ruled out. Through the identical panel construction its tail exponent never exceeds 2.63.
3. The vendor's kernel accounts for the steepness of the tail but not its shape. Smoothed diffusion matches the exponent and fails on the profile.
4. Between 43% and 68% of the variance of the second factor moves with the gap between call and put volatilities at the money, which the vendor's separate smoothing of calls and puts can produce.
5. Several modes are negatively autocorrelated over two to five days, a pattern that measurement noise alone does not create. Its source is not identified.

Section 2 sets out the membrane model and the prediction for daily changes. Section 3 describes the data, the panel and the vendor's kernel, and Sec. 4 the checks on the code. Sections 5 to 8 report the measurements and tests, and Sec. 9 discusses what the instrument can and cannot decide. The decision rules for the tests were written down before the tests were run; Appendix A records them together with the one rule that was amended after its results existed. This version replaces an earlier draft whose results came from a flawed panel; Appendix B lists the corrections.

2 The membrane model and its prediction for daily changes

On a moneyness window $[z_{\min}, z_{\max}]$ of length L with reflecting (Neumann) boundaries, the field decomposes onto cosine modes $\varphi_k(z) = \sqrt{2/L} \cos(q_k(z - z_{\min}))$ with wavenumbers $q_k = k\pi/L$. Each mode amplitude follows an Ornstein Uhlenbeck (OU) process $\dot{f}_k = -\gamma_k f_k + \xi_k$ with stationary variance $V_k = \sigma^2/(2\gamma_k)$. I allow a bending term alongside diffusion,

$$\gamma_k = D q_k^2 + \kappa q_k^4, \quad (2)$$

so that $\kappa = 0$ is pure diffusion and D and κ have units of z^2 and z^4 per day.

Implied volatility levels are close to a random walk, so the analysis uses daily changes. For a mode sampled at interval dt , write $a = e^{-\gamma dt}$. The exact OU transition is $f(t + dt) = a f(t) + \eta$ with η independent of $f(t)$ and $\text{Var}(\eta) = V(1 - a^2)$, so

$$\text{Var}(\Delta f) = V(1 - a)^2 + V(1 - a^2) = 2V(1 - e^{-\gamma dt}). \quad (3)$$

For slow modes ($\gamma dt \ll 1$) this tends to $\sigma^2 dt$, independent of k . For fast modes it tends to $2V \propto \gamma^{-1}$. A spectrum of daily changes is therefore flatter than the familiar k^{-2} law for levels, and the steepest it can be under pure diffusion is the fast limit.

The same transition fixes the autocorrelation of daily changes, which Sec. 8 uses. For lag $h \geq 1$,

$$\text{Corr}(\Delta f_t, \Delta f_{t+h}) = -\frac{1}{2}(1 - a)a^{h-1}. \quad (4)$$

Independent measurement noise added to the levels also produces a lag one correlation of $-\frac{1}{2}$ in the changes, but it leaves all longer lags at zero. Negative correlation at lags of two days or more therefore points to relaxation on a scale of days. Relaxation much faster than one day ($a \rightarrow 0$) cannot be distinguished from noise at daily sampling.

Daily changes of the real smile are dominated by a component that moves the whole smile together. I model it as a separate level factor that adds a uniform variance m to every point, so the covariance of daily changes at points z_i and z_j is

$$\text{Cov}(\Delta p_i, \Delta p_j) = \sum_k \varphi_k(z_i) \varphi_k(z_j) 2V_k(1 - e^{-\gamma_k dt}) + m. \quad (5)$$

Normalising this to a correlation matrix and diagonalising it gives the variance shares the model predicts for the panel. I summarise a spectrum by its tail exponent p , the slope of a least squares fit of $\log \lambda_k$ against $\log k$ over modes $k = 2, \dots, 7$, and by the local slopes between neighbouring modes.

3 Data, panel and the vendor's kernel

3.1 Data

The data are OptionMetrics IvyDB standardised volatility surfaces for SPX (table `vsurfd`, secid 108105) from 2 January 2015 to 31 December 2024, with the index level from `secprd`. The surface gives implied volatility at fixed maturities and fixed deltas, together with the strike that the vendor's own model implies for each pillar. I use the 30 calendar day maturity.

3.2 Moneyness and panel construction

Each pillar is placed at

$$z = \frac{\log(K/S)}{\sigma_{\text{ATM}}\sqrt{T}}, \quad (6)$$

where K is the vendor's implied strike, S the index level, T the maturity in years and σ_{ATM} the average of the 50 delta call and put volatilities on that day. Using the vendor's strike keeps the coordinate consistent with the vendor's own pricing assumptions. Scaling by σ_{ATM} rather than by each pillar's own volatility avoids a coordinate that merely relabels delta. Calendar days are converted to the 252 day trading year before computing T .

Only out of the money pillars are kept ($|\Delta| \leq 0.50$), so puts populate $z < 0$ and calls $z > 0$ and each side of the smile comes from its more liquid instrument. Pillars are averaged within 8 equal bins on $z \in [-1.15, 1.10]$, the window the delta grid can fill. A bin is kept if it has data on at least 95% of days, and days with any empty kept bin are then dropped. The deepest put bin has data on only 89.1% of days and is dropped. The panel is 2498 days by 7 bins, with bin centres at $z = -0.728, -0.447, -0.166, 0.116, 0.397, 0.678, 0.959$. The analysis variable is the daily change in implied volatility. It differs from delta hedged option returns by the gamma and theta carry, which mainly affects the level factor; the interpretation of mode 1 carries that caveat.

3.3 The vendor's kernel

According to the IvyDB reference manual [3], the volatility at each pillar j is a weighted average of the implied volatilities σ_i of all options on that day,

$$\hat{\sigma}_j = \frac{\sum_i V_i \Phi(x_{ij}, y_{ij}, z_{ij}) \sigma_i}{\sum_i V_i \Phi(x_{ij}, y_{ij}, z_{ij})}, \quad \Phi \propto \exp\left(-\frac{x^2}{2h_1} - \frac{y^2}{2h_2} - \frac{z^2}{2h_3}\right), \quad (7)$$

where V_i is the option's vega, $x_{ij} = \log(T_i/T_j)$, y_{ij} is the difference in call equivalent delta (delta for a call, one plus delta for a put) and z_{ij} is 0 if the two are the same type and 1 otherwise. The bandwidths are $h_1 = 0.05$, $h_2 = 0.005$ and $h_3 = 0.001$, and options with vega below 0.5 are excluded.

Two consequences matter here. First, $\sqrt{h_2} \approx 0.07$ in delta is wider than the 0.05 spacing between pillars, so neighbouring pillars share most of their inputs. Second, the weight between a call and a put is e^{-500} , so the two halves of the surface are built from disjoint sets of options. Because of the stitching rule above, the put side of the panel comes entirely from the put surface and the call side from the call surface, and the two meet at the money. The kernel is therefore part of the measurement. Any statement about the dynamics of the smile has to pass through it.

Table 1: Variance shares of the principal components of daily changes (2497 changes, 7 bins). The tail columns give each mode’s share of the variance of modes 2 to 7 combined. Brackets are 95% bootstrap intervals.

Mode	Share (%)	95% interval	Share of tail	95% interval
1	94.37	[93.59, 95.17]		
2	4.71	[3.94, 5.46]	0.837	[0.798, 0.865]
3	0.63	[0.49, 0.82]	0.112	[0.089, 0.148]
4	0.17	[0.13, 0.21]	0.030	[0.023, 0.039]
5	0.07	[0.05, 0.08]	0.013	[0.010, 0.014]
6	0.03	[0.02, 0.04]	0.006	[0.004, 0.008]
7	0.02	[0.01, 0.02]	0.003	[0.002, 0.004]

Table 2: Loadings of the first three modes on the seven bins. Signs are fixed so that mode 1 has a positive sum, mode 2 is positive at the call side edge and mode 3 is positive at its largest loading.

Bin centre z	−0.728	−0.447	−0.166	0.116	0.397	0.678	0.959
Mode 1	0.377	0.378	0.376	0.386	0.380	0.377	0.373
Mode 2	−0.403	−0.414	−0.423	0.043	0.330	0.424	0.444
Mode 3	−0.039	0.078	0.279	−0.522	−0.508	0.131	0.605

4 Validation

The simulator evolves each mode with the exact OU transition, so there is no time discretisation error, and adds an optional level factor. A unit test simulates 20,000 days on a 50 point grid and checks that PCA of the daily changes recovers the variance shares of Eq. (3) to within 1.2 percentage points per mode. A second test checks the analytic covariance of Eq. (5), evaluated at the panel’s bin centres with 40 modes, against direct simulation; the four leading shares agree to within 10%. Agreement on synthetic data checks the code. It is not evidence about the market.

5 The measured spectrum

Uncertainties throughout come from a moving block bootstrap of the daily changes (blocks of 25 days, 1000 replicates), which preserves volatility clustering.

Table 1 gives the spectrum and Table 2 the leading mode shapes. Mode 1 is almost exactly uniform and is the level of the smile. Mode 2 is not the smooth tilt a membrane would produce. It is roughly constant across the three put side bins, roughly constant across the three call side bins, and changes sign in the bin that contains the at the money pillars, where the two surfaces meet. Mode 3 concentrates on the two bins just above the money ($z = 0.116$ and 0.397) and on the outermost call bin, and has no clean harmonic form.

The tail exponent over modes 2 to 7 is $p = 4.44$, with interval 4.18 to 4.67. The local slopes between neighbouring modes are 4.97, 4.54, 3.94, 4.42 and 3.72, with intervals [4.16, 5.63], [3.47, 5.62], [2.67, 5.56], [1.61, 6.32] and [2.43, 5.54]. None of them is distinguishable from 4.4, so on log scales the measured tail is consistent with a straight line.

Table 3: Tail exponent p of synthetic pure diffusion through the study panel, mean over three seeds (seed standard deviation at most 0.1). The measured value is 4.44 with interval [4.18, 4.67].

D (z^2/day)	Kernel off			Kernel on		
	$q = 0$	$q = 0.2$	$q = 0.5$	$q = 0$	$q = 0.2$	$q = 0.5$
0.003	1.05	0.85	0.60	3.93	3.79	3.48
0.01	1.08	0.91	0.62	4.12	3.98	3.65
0.03	1.28	0.98	0.67	4.49	4.31	3.88
0.1	1.79	1.19	0.76	5.09	4.84	4.24
0.3	2.22	1.37	0.83	5.63	5.29	4.53
1	2.55	1.48	0.88	6.01	5.58	4.70
3	2.63	1.49	0.89	6.09	5.63	4.73

6 Unsmoothed diffusion and the vendor’s kernel

6.1 Design of the test

The test asks whether the kernel alone can turn a pure diffusion membrane ($\kappa = 0$) into the measured tail. Synthetic implied volatilities follow $\sigma(z, t) = (\sigma_0 + \ell(t)) s(z) + p(z, t) + \epsilon$, with a fixed smile shape $s(z) = 1 - 0.30z + 0.08z^2$ and $\sigma_0 = 0.16$. The membrane p lives on the study window with 40 modes and is scaled so that its daily change has a standard deviation of 0.25 volatility points. The level ℓ is a slow OU process with daily kick 0.0075, and ϵ is independent quote noise with standard deviation q . Quotes are generated for calls and puts at expiries of 16, 23, 30, 37 and 44 days and at z from -4 to 3 in steps of 0.05. The vendor’s kernel turns them into 30 day pillars at the IvyDB deltas, the pillar strike is recovered from its smoothed volatility, and the result goes through exactly the same panel construction as the real data, restricted to the same seven bins. With the kernel switched off, each pillar takes the nearest quote instead.

The grid is $D \in \{0.003, 0.01, 0.03, 0.1, 0.3, 1, 3\}$ in z^2 per day and $q \in \{0, 0.2, 0.5\}$ volatility points, with three seeds of 2500 days per cell. The level kick was chosen before the run so that mode 1 carries about 94% at one reference cell, so the agreement in mode 1 is tuned by construction. No tail quantity was examined before the run.

6.2 Results

Table 3 shows the result. Without the kernel the tail exponent saturates at 2.630 ± 0.013 in the fast limit, well below the measured 4.44. Unsmoothed diffusion is therefore ruled out on this instrument. An earlier version of this analysis put the ceiling at 1.87. The ceiling has risen, but the measured tail has risen more, so the gap between them has grown from 0.66 to 1.81. An analytic decomposition, applying one change at a time, starts from 1.86 on the earlier instrument. Sampling a continuous field instead of a truncated one adds 0.17, sampling at the bin centres adds 0.10, dropping the deepest put bin subtracts 0.20, adding the level factor adds 0.46, and the pillar and binning steps of the full simulation add the last 0.23. The level factor is the largest single contribution, although the split depends on the order in which the changes are applied.

With the kernel switched on the exponent rises by between 2.9 and 4.1 in every cell. Four cells land inside the measured interval: 4.49 ± 0.03 , 4.31 ± 0.02 , 4.24 ± 0.01 and 4.53 ± 0.02 . Since the runs with and without the kernel differ only in the bandwidths, the entire shift from 2.63 to about 4.4 is attributable to the kernel.

The shape is a different matter (Fig. 1). In the four matching cells, mode 2 holds between 0.45 and 0.57 of the tail variance against a measured 0.837 [0.798, 0.865], and none of the six tail shares lies inside its measured interval in any cell. The local slopes rise steeply, from between

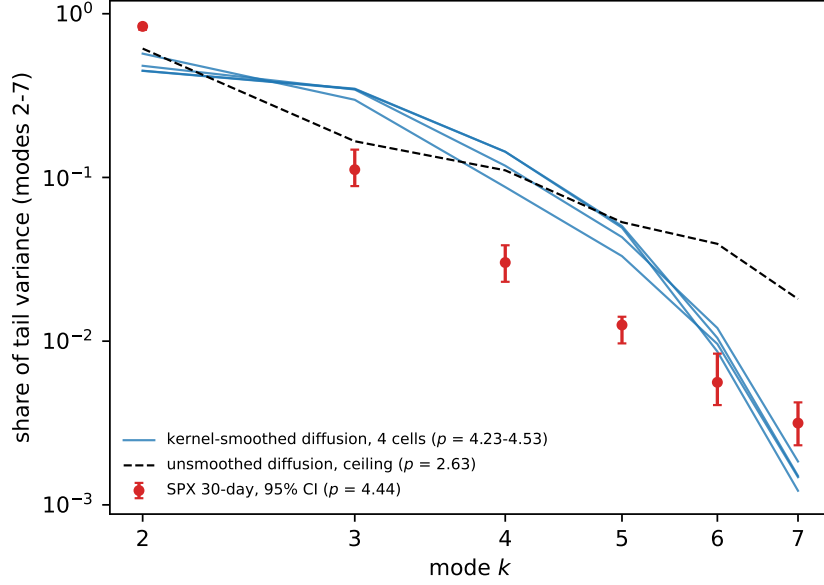


Figure 1: Shares of tail variance (modes 2 to 7) on log scales. Red: the SPX 30 day panel with 95% bootstrap intervals. Blue: kernel smoothed pure diffusion in the four cells whose exponent lies inside the measured interval. Dashed: unsmoothed diffusion at its ceiling. The smoothed spectra match the measured exponent but bend downward, while the measured tail stays close to a straight line.

Table 4: Share of each factor’s daily return variance explained by the daily change in the call minus put volatility at the money (R^2), with 95% block bootstrap intervals.

Mode	R^2	95% interval
1	0.026	[0.009, 0.047]
2	0.572	[0.427, 0.678]
3	0.204	[0.051, 0.441]
4	0.039	[0.011, 0.088]

0.6 and 1.6 at the first step to between 12.2 and 12.7 at the last, and only the middle one or two of the five fall inside the measured intervals. The same rising pattern appears in every one of the 21 cells with the kernel switched on. It is what Gaussian smoothing does to a spectrum: it suppresses the high modes progressively harder. The measured spectrum does not show it. The level factor is not responsible for the mismatch, because these comparisons use the tail alone.

The vendor’s kernel can therefore produce the steepness of the measured tail but not the shape of that steepness. On this evidence the steep tail cannot be credited to a bending term in the dynamics, and it cannot be dismissed as smoothing either.

7 The call and put seam

The step in mode 2 sits exactly where the panel switches from the put surface to the call surface. If the two surfaces carry independent errors, their difference at the money should move from day to day. I measure that difference directly, $\delta_t = \sigma_{50}^{\text{call}}(t) - \sigma_{50}^{\text{put}}(t)$ at 30 days, and regress the daily returns of each factor on $\Delta\delta_t$. For a single consistent surface the two 50 delta volatilities refer to the same strike and δ_t would not move.

The rule fixed in advance reads mode 2 as not substantially tied to the seam if the upper end of its interval is below 0.5, as substantially tied if the lower end is above 0.5, and as ambiguous

Table 5: Autocorrelation of daily factor returns at lag 1 and the sum S of the autocorrelations at lags 2 to 5, with 95% block bootstrap intervals.

Mode	Lag 1	95% interval	S (lags 2 to 5)	95% interval
1	-0.098	$[-0.153, -0.041]$	-0.052	$[-0.156, 0.055]$
2	-0.295	$[-0.365, -0.192]$	-0.170	$[-0.309, -0.007]$
3	-0.279	$[-0.379, -0.155]$	-0.069	$[-0.135, 0.014]$
4	-0.269	$[-0.338, -0.192]$	-0.105	$[-0.165, -0.007]$
5	-0.287	$[-0.386, -0.187]$	-0.119	$[-0.197, -0.016]$
6	-0.385	$[-0.422, -0.312]$	-0.067	$[-0.128, 0.004]$
7	-0.336	$[-0.404, -0.241]$	-0.115	$[-0.211, -0.012]$

otherwise. Table 4 gives $R^2 = 0.572$ with interval 0.427 to 0.678, so the formal reading is ambiguous. Even the lower end of that interval means that more than two fifths of the second factor's daily variance moves with a quantity that should not move at all for a single surface. Mode 3 is affected to a lesser degree, and modes 1 and 4 hardly at all. The second factor of this panel, conventionally read as skew, is therefore partly a property of how the vendor assembles the surface.

8 Relaxation over a few days

This test is a measurement and does not use any fitted rates. By Eq. (4), a negative lag one autocorrelation on its own cannot separate relaxation from measurement noise, so the criterion fixed in advance also requires longer lags. It is met if the lag one interval lies below zero for modes 2 and 3 and the interval for S lies below zero for at least one of them. On simulated panels from the harness of Sec. 6, a membrane relaxing much faster than one day gives lag one values near -0.5 with S near zero and fails the criterion, while $\gamma_1 \approx 0.2$ per day gives lag one near -0.25 with S near -0.13 and passes.

Table 5 shows that all tail modes have lag one autocorrelations between -0.27 and -0.39 , and that S is significantly negative for modes 2, 4, 5 and 7. The criterion is met, so there is structure on a scale of days that noise alone does not produce. Its source is open. Mode 2 is substantially tied to the seam (Sec. 7), and vendor errors that decay over a few days would leave exactly this signature. Pillars drifting between bins as the smile moves could do the same in the higher modes. Relaxation faster than about one day is invisible at daily sampling whatever its source, so this test cannot rule out a fast operator.

9 Discussion

What the data show. The 30 day SPX smile, as published by the vendor, moves mostly as a whole, with a second factor that changes sign at the money and a tail that falls as $k^{-4.4}$. Diffusion observed without smoothing cannot produce that tail. Diffusion passed through the vendor's kernel reproduces its steepness but not its shape. The second factor is partly tied to the seam between the call and put surfaces, and several factors relax over a few days for reasons that are not yet clear.

What the instrument cannot decide. Before tests 47 and 48 were run, I fixed a rule for what would follow. A forward model that includes the kernel, fitted to the measured spectrum, would be built only if mode 2 proved not to be substantially tied to the seam and the relaxation criterion was met. The seam result was ambiguous, so the rule was not satisfied and that model

was not built. The question of which operator, if any, governs the smile therefore remains open. It has not been answered in either direction.

The measurement that would decide it. Rebuilding the 30 day slice from raw option quotes (OptionMetrics `opprcd`), as one surface consistent with parity between calls and puts, removes both the kernel and the seam. A spectrum measured that way could be compared with Eq. (5) directly. The same data would also reach deeper into the wings than the delta grid allows and give access to higher modes.

Limitations. The synthetic harness has no structure across maturities, so the maturity part of the kernel acts on a field that is the same at every expiry; it does not model the vega cutoff; and its quote noise is independent across options. The analysis uses one maturity and one index. Changes in implied volatility at fixed z include the gamma and theta carry, which mainly affects mode 1. The tail statistics rest on six modes, and the smallest shares are of order 10^{-4} .

Acknowledgements

This project was inspired by a conference talk by P. Ioselevich on factor models for option risk. The analysis here is independent and differs substantially from that work.

Use of AI tools

I used AI assistants (mainly Claude, through chat and Claude Code) throughout this project. They wrote a large share of the code from my specifications and helped me learn the methods involved. I also used an AI model as a critical reviewer of the earlier draft; that review found the errors listed in Appendix B. The research question, the choice of data, the decisions about what to test and what to claim, and the interpretation are mine, and every number in this paper comes from scripts I ran myself.

A Prediction ledger

Each test states its decision rule in the docstring of its script, written before the test was run. The entries below record the rules and their outcomes, in the order they happened.

Test 49, validity check. The rule required that unsmoothed diffusion give a tail exponent of at most 2.5, so that the harness could not produce the measured tail without the kernel. The threshold was derived from the diffusion ceiling of the earlier instrument (1.87), which no longer applied after the corrections in Appendix B. The check failed: the ceiling on the corrected instrument is 2.630 ± 0.013 . After these results existed, the check was amended to test its stated purpose directly: the largest exponent without the kernel, plus two seed standard deviations, must lie below the lower end of the measured interval. The amended check passes ($2.656 < 4.182$). The conclusions in Sec. 6 rest on the results with the kernel switched on, not on this check.

Test 49, verdict. Smoothing is sufficient for the tail exponent: four cells with the kernel on lie inside the measured interval. It fails on the shape, inside the tail: mode 2 holds 0.45 to 0.57 of the tail variance against a measured 0.837, and the local slopes rise from about 1 to about 12.5 against a measured profile that stays near 4.4. The level kick was fixed before the run by trying three values at one reference cell and looking only at the mode 1 share, so the agreement in mode 1 is tuned. The operator question stays open.

Test 47. $R^2 = 0.572$ [0.427, 0.678] for mode 2. Ambiguous under the rule, since the interval contains 0.5.

Test 48. Criterion met: lag one negative for all tail modes, S negative for modes 2, 4, 5 and 7. The test detects relaxation on scales of a day or longer only, so failure would have closed the question of dynamics at daily sampling without deciding anything about the static structure. The source of the relaxation is not identified.

Scope rule. Fixed before tests 47 and 48 reported: build a forward model that includes the kernel only if test 47 gives an upper interval end below 0.5 and test 48 meets its criterion. Test 47 did not, so the model was not built, although test 48 passed. The fitted membrane model of the earlier draft and the out of sample risk comparison built on it were removed rather than refitted, since the paper no longer makes claims that depend on them.

Measurement model. The vendor's kernel is treated as part of the measurement, not as a limitation. Any claim about an operator, in this work or in an extension to finer data, needs a forward model that includes it.

B Corrections to the earlier draft

An earlier draft reported a fitted membrane with tension and bending, a feature in modes 4 and 5 that the fit could not reproduce, and an out of sample risk comparison. A review of that draft found the following problems, all corrected here.

1. The analysis scripts averaged implied volatility within each z bin across all eleven vendor maturities instead of using the 30 day slice. The maturity filter was present in the script that built the reference panel and missing from the others. Every result of the earlier draft came from the averaged panel, including the feature in modes 4 and 5, which does not appear in the 30 day slice. The panel is now built in a single function that checks that it contains one maturity.
2. The earlier draft attributed a dropped bin to a problem with data types. The real cause was the missing filter: the 30 day slice genuinely lacks the deepest put bin on 10.9% of days.
3. The decay rates used integer mode numbers instead of wavenumbers, so D and κ were not in units of z . The model was evaluated at evenly spaced points instead of at the bin centres, and it truncated the field to as many modes as sample points, which made one mode an artifact of the truncation.
4. The out of sample comparison used model parameters fitted on the full sample, test period included, although the draft described all models as built from training data only.
5. The draft stated that a butterfly portfolio loads on modes 3 to 5. Its symmetric weights give it almost no exposure to mode 4, and under the fitted model almost all of its variance sat in mode 3.
6. The composite fit had $\chi^2 = 14.4$ on 4 degrees of freedom ($p \approx 0.006$) on the averaged panel, which the draft described as a model that works. That fit is recorded here as history only and is not a property of the 30 day panel.

C Reproducibility

The code is at the repository above. A script downloads the data once from WRDS; the data are licensed and not included. Every number in this paper is printed by one of the following scripts: `build_panel.py` (panel, spectrum, loadings), `bootstrap_real_spectrum.py` (all intervals in Sec. 5), `test49_vendor_kernel.py` (Table 3), `test49_followups.py` (the ceiling decomposition and the shape comparison), `test47_48.py` (Tables 4 and 5, and in its validation mode the criterion check on simulated panels) and `make_figures.py` (Fig. 1). Randomness is seeded throughout.

References

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